nature portfolio

Peer Review File

Utilizing a Publicly Accessible Automated Machine Learning Platform to Enable Diagnosis before Tumor Surgery

Corresponding Author: Professor Zara Patel

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Attachments originally included by the reviewers as part of their assessment can be found at the end of this file.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

**Review of "Utilizing a Publicly Accessible Automated Machine Learning Platform
2 to Enable Diagnosis before Tumor Surgery" **

The paper offers an analysis of the effectiveness of an AI-assisted imaging tool, focusing on its use of Google's Vertex AI and pre-operative CT scans to accurately detect malignant transformation of benign inverted papillomas (IP). The authors claim that the AutoML model can assist physicians with limited coding experience in distinguishing benign from malignant tumors more accurately than previously published methods. These findings introduce a novel approach that could reduce unnecessary head and neck surgeries, potentially impacting the decision-making process of Otolaryngology-Head and Neck Surgeons when handling potential cases of IP or IP-SCC.

The paper's innovation lies in integrating an AI algorithm into conventional preoperative diagnostic imaging. This research is particularly pertinent to ENT surgeons, as it addresses the challenges of obtaining reliable diagnostic samples in complex areas like the sinonasal tract and skull base.

However, while the claims made in the paper are compelling, the value of the results would be enhanced by clarifying whether the final diagnosis of IP or IP-SCC for the study cases was confirmed through biopsies or resections. What are the concordance rates between the AutoML model, biopsies, and resections? Additionally, what role would AutoML play in cases where diagnostic biopsies are available to inform subsequent surgical decisions? What key imaging features did the AI learn to distinguish between IP, IP-SCC, and other diagnoses?

In terms of impact, the paper is likely to influence both ENT surgeons and pathologists, particularly in how treatment decisions are made without relying on traditional pathological diagnoses. The innovative nature of the findings could also open the door for future research on AI-assisted imaging in the diagnosis of other sinonasal and skull base tumors.

Reviewer #2

(Remarks to the Author)
The manuscript "Utilizing a Publicly Accessible Automated Machine Learning Platform to Enable Diagnostic Before Tumor Surgery" presents an innovative research proposal that aims to use a publicly available platform, Google Vertex AI, and pre-operative CT imaging of patients from nineteen 116 separate institutions with IP or IP-SCC to identify malignant transformation of a benign tumor. The authors' choice of this engaging topic and innovative research proposal is highly commendable. However, to further enhance the manuscript's potential for publication, the following suggestions should be considered:
1. The authors should review the keywords using MeSH terms. If not, please replace them as necessary.
2. The authors are encouraged to ensure the consulted references are correctly added in the introduction section, per the journal's guidelines. They should also carefully explain the meaning of each abbreviation used in the manuscript.
3. The authors might modify lines 160 to 163 from page 6 from "Our study aims to harness AI technology via an automated

machine learning (AutoML) algorithm to develop a prediction model to differentiate between IP and IP-SCC, with an eventual aspiration to raise the level of diagnosis and treatment of these tumors, and many more benign tumors with the same potential for malignant transformation, in underserved areas around the globe" to "Our study aims to harness AI technology via an automated machine learning (AutoML) algorithm to develop a prediction model to differentiate between IP and IP-SCC to increase the accurate diagnosis and treatment of these lesions."
4. The authors should list the names of the 19 institutions worldwide participating in the study in a Table or supplemental data. They should also provide the number of IRB-approved protocols in that file. In addition, the authors should incorporate lines 166 to 168 at the end of the dataset section. Then, they should add the number of patients from each academic center and establish the inclusion and exclusión criteria followed.
5. The authors should quote the applied references for histopathology diagnosis, image processing, and model training.
6. The autos should add the algorithm and code used for the analysis in the methods section.
7. The authors should set a statistical analysis section.
8. The authors should organize the CT images in Figure 1 into coronal, sagittal, and axial views.
9. The authors should briefly explain Figures 2 and 3.
10. The authors are encouraged to start the discussion section with a paragraph summarizing the study's most relevant findings. This will help the authors to realize the significant impact of their work. Subsequently, the authors should compare these outcomes with similar studies worldwide to further emphasize the importance of their findings. The authors should change the syntax from lines 236 to 238 on page 9 since it is difficult to follow. In addition, the authors should go deep into how diversity strengthens their research, which seems a contradictory idea because it is set as a limitation in a posterior paragraph.

Reviewer #3

(Remarks to the Author)
This paper explores the use of a publicly available deep-learning platform to distinguish between benign and malignant tumors in head CT scans, specifically focusing on the classification of sinonasal inverted papilloma versus squamous cell carcinoma.

While the methodology itself is not novel, as it relies on an existing deep-learning platform, its direct application to this specific clinical problem is noteworthy. The implementation of deep learning in this context has the potential to be transformative for clinicians, offering a powerful tool for diagnostic assistance. However, the paper falls short in several key areas.

The dataset description lacks sufficient detail, making it difficult to assess the quality and diversity of the data used for model training. Additionally, the image preprocessing steps are inadequately explained, leaving uncertainty about how the data was prepared and standardized. The model training process is described in a rather superficial manner, without enough depth to evaluate its robustness or reproducibility.

In contrast, the discussion section is well-written and provides valuable insights. However, an important issue—the high false negative rate—is not adequately addressed. This is a critical concern in clinical applications, where misclassifying malignant tumors as benign could have serious consequences. A more thorough analysis of this limitation, along with potential strategies to mitigate it, would have strengthened the paper significantly.

Moreover, while the study highlights the potential impact of deep learning in clinical decision-making, it would have been beneficial to include a discussion on interpretability and explainability. These aspects are crucial for clinician adoption, as they provide transparency into the model's decision-making process and build trust in AI-assisted diagnoses.

Overall, this paper presents an interesting application of deep learning in medical imaging, but it requires more rigor in dataset description, model training details, and a deeper discussion of critical results to be fully convincing. Future iterations could greatly benefit from addressing these gaps and integrating explainability methods to enhance clinical usability.

Reviewer #4

(Remarks to the Author)
The manuscript presents a multi-institutional study that leverages Google Vertex AI, a publicly accessible automated machine learning platform, to develop a model for differentiating between benign sinonasal inverted papilloma (IP) and its malignant transformation (IP-SCC) using pre-operative CT imaging. Drawing from a dataset of 958 patients (878 with IP, 80 with IP-SCC) across 19 international institutions, the authors created an automated machine learning model that achieved impressive results: 99.8% AUC, 99.2% precision, 95.8% sensitivity, 99.7% specificity, and 99.1% overall accuracy in distinguishing between benign and malignant tumors without requiring specialized AI expertise.

Some positive aspects:
-The study demonstrates exceptional clinical value by addressing a significant diagnostic challenge in tumor surgery. The ability to accurately differentiate between benign IP and malignant IP-SCC preoperatively has important implications for surgical planning and patient counseling. The model's high accuracy (99.1%) surpasses previously published rates from human experts, suggesting potential for real clinical impact in reducing reliance on invasive biopsies that may be subject to

sampling error.

-The international multi-institutional nature of the study (19 centers worldwide) and large dataset (41,099 CT scan cuts from 958 patients) represent significant methodological strengths. The diversity in patient demographics, imaging equipment, and scanning parameters enhances the model's generalizability across different clinical settings. Unlike studies using standardized or homogeneous datasets, this approach better mirrors real-world conditions where image quality and acquisition parameters vary.

-The authors effectively demonstrate that non-AI specialists can leverage publicly accessible AutoML platforms to develop highly accurate diagnostic models with minimal coding requirements. This democratization of AI tool development is particularly valuable for the medical community, as it suggests that clinicians without extensive programming skills can still harness the power of AI for diagnosis, potentially accelerating clinical implementation and adoption.

Some shortcomings of the manuscript:
-The study population exhibits a significant class imbalance between benign IP (n=878) and malignant IP-SCC (n=80) cases. While the authors acknowledge this limitation, they could more thoroughly discuss how this imbalance might impact model performance metrics, particularly sensitivity for detecting the minority class. A more detailed explanation of techniques employed to mitigate this imbalance would strengthen the methodology.

-The manuscript lacks sufficient technical details about the specific algorithms employed by the Google Vertex AI platform. While the authors highlight the ease of use of AutoML, a deeper explanation of the underlying architecture, feature extraction methods, and model selection processes would enhance reproducibility and provide more transparency about the "black box" nature of the system.

-The retrospective design introduces potential selection bias. The authors should provide more information about how patients were selected at each institution and discuss whether cases were consecutive or selected based on specific criteria, which could impact the generalizability of the results.

-The validation methodology appears to rely on internal validation (splitting the existing dataset), rather than true external validation on an entirely separate cohort. Given the multi-institutional nature of the study, the authors missed an opportunity to perform more rigorous validation by holding out entire institutions for testing rather than just randomly selected images.

-The manuscript does not adequately address potential confounding variables that might influence model performance. For instance, tumor size, location, calcification patterns, or other radiological features could serve as spurious correlates that the model might exploit rather than true discriminative features of malignancy.

-It is very important for the manuscript to propose that doctors with few engineering technical backgrounds use publicly available AI tools, and the related methods are also worth studying. It is suggested that the author can cite more similar papers, such as: ：AI-Generated Content Enhanced Computer-Aided Diagnosis Model for Thyroid Nodules: A ChatGPT-Style Assistant, arXiv, 2024, doi.org/10.48550/arXiv.2402.02401

-While the authors briefly mention affordability challenges of hosting the model on Google Vertex Cloud, they should expand on the practical implementation barriers. A more detailed cost-benefit analysis and discussion of alternative deployment strategies would enhance the translational aspect of the research.

-The paper would benefit from a more thorough comparison with existing radiological criteria used by human experts to distinguish IP from IP-SCC. Including side-by-side performance metrics of radiologists using established criteria versus the AI model on the same test set would provide stronger evidence for the superior performance claimed by the authors.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
One of my initial concerns pertained to the concordance between the AI model, biopsy findings, and resection specimens. The authors responded that they did not collect data from pre-resection biopsies and instead focused solely on comparing AI predictions with the final pathology of resection specimens. However, this appears inconsistent with the methodology described in the paper, where the dataset is said to include biopsy-proven cases of IP versus IPSCC. These biopsy cases were reportedly used to train the AI model.
This raises important questions: if the biopsies did not show strong concordance with the final resection diagnoses, are they truly reliable for training an AI model? Conversely, if they were concordant, then we already have a reliable diagnostic tool in the form of biopsy pathology—prompting the question of what additional value the AI model brings.
These conflicting elements make it perplex to interpret the reported high concordance between the AI model and final pathology within this internal validation. A prospective external validation study would be crucial to substantiate these findings.
In clinical practice, discrepancies between biopsy and resection diagnoses for tumors such as IP and IPSCC often arise

from sampling limitations rather than histopathological interpretation. If AI can predict the likelihood of malignancy from imaging when interpreted by a low coding experience clinicians, the discussion of the paper can also focus on its other practical application which may lie in guiding targeted biopsies—helping clinicians obtain more representative samples—rather than functioning as a standalone diagnostic tool.

Reviewer #2

(Remarks to the Author)
I appreciate the authors' effort and kindness in following the suggested modifications to their manuscript. The authors share an engaging and innovative research topic that is highly commendable. Thus, the manuscript is optimal for publication.

Reviewer #3

(Remarks to the Author)
The paper has improved due to the comments and the authors have made significant efforts to address all comments.

Reviewer #4

(Remarks to the Author)
I believe the authors have provided comprehensive responses to my previous review. They've effectively addressed key concerns regarding class imbalance, technical transparency, validation methodology, and implementation barriers. The manuscript now more clearly acknowledges limitations while strengthening its clinical relevance. Their explanations about mitigating selection bias and confounding variables are particularly satisfactory. While direct comparison with radiologist performance remains a future direction, the authors have outlined clear plans to address this through explainable AI techniques and prospective studies. The revisions have significantly enhanced the manuscript's methodological clarity and translational value, making a compelling case for the utility of accessible AI tools in clinical diagnosis.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Thank you for the clarification of methodology and terminology. I think the author has diligently answered my questions.

Dear Professor Salvatico and esteemed reviewers,

Thank you for the expert and extensive review of our manuscript. We have significantly revised the manuscript based on your comments and suggestions, and we think the submission is stronger because of it.

Please find below our point by point response to each comment.
We hope these answers appropriately satisfy your concerns.

Sincerely,

Zara M. Patel, on behalf of all authors

Reviewer #1 (Remarks to the Author):

**Review of "Utilizing a Publicly Accessible Automated Machine Learning Platform to Enable Diagnosis before Tumor Surgery" **

The paper offers an analysis of the effectiveness of an AI-assisted imaging tool, focusing on its use of Google's Vertex AI and pre-operative CT scans to accurately detect malignant transformation of benign inverted papillomas (IP). The authors claim that the AutoML model can assist physicians with limited coding experience in distinguishing benign from malignant tumors more accurately than previously published methods. These findings introduce a novel approach that could reduce unnecessary head and neck surgeries, potentially impacting the decision-making process of Otolaryngology-Head and Neck Surgeons when handling potential cases of IP or IP-SCC.

The paper’s innovation lies in integrating an AI algorithm into conventional preoperative diagnostic imaging. This research is particularly pertinent to ENT surgeons, as it addresses the challenges of obtaining reliable diagnostic samples in complex areas like the sinonasal tract and skull base.

However, while the claims made in the paper are compelling, the value of the results would be enhanced by clarifying whether the final diagnosis of IP or IP-SCC for the study cases was confirmed through biopsies or resections.

What are the concordance rates between the AutoML model, biopsies, and resections?

One of the main underlying reasons why pre-operative imaging is so important in this disease pathology, is that while sometimes the tumor is accessible for biopsy in the office – with the anterior aspect of the tumor sometimes not showing malignant change that is present deeper in the tumor – but also that an in-office biopsy is not even an option for many of these tumors that are situated more deeply within the sinus cavities. In those situations, a surgical resection is planned and it is only intra-operatively that we sometimes realize there has been malignant change and then have to stop, wake the patient, and plan for an entirely different surgery at a later date, while many others are able to move forward to completion at that same setting. Due to this heterogeneity in whether patients have a separate biopsy versus simply one surgical resection, we did not try and collect pre-resection biopsy results. The AutoML model was trained and validated using CT images labeled according to final resection pathology results, making the reported performance metrics — including sensitivity (95.8%), specificity (99.7%), and overall accuracy (99.1%) — reflective of the concordance between the AutoML model and final pathology findings. Since the final pathology results served as the ground truth in this model, these metrics represent the concordance rate between the AutoML predictions and pathology-confirmed final diagnoses.

Additionally, what role would AutoML play in cases where diagnostic biopsies are available to inform subsequent surgical decisions?

Thank you for this important question. We have added this response to the discussion section.

As noted above, pre-operative biopsies are not always an option for these patients, but even when diagnostic biopsies are available, AutoML has a meaningful complementary and potentially transformative role in informing surgical planning.

While pre-operative biopsies remain a standard diagnostic tool, they can be limited by sampling error, especially in tumors with heterogeneous pathology like inverted papilloma (IP) that can undergo focal malignant transformation to IP-SCC. A biopsy might miss malignant regions, leading to misdiagnosis and affecting decisions regarding the extent of surgical resection.

AutoML offers a noninvasive, image-based evaluation of the entire tumor volume, not restricted to a specific sampled area. This broader perspective can identify radiographic features suggestive of malignancy that biopsies may miss. As such, AutoML would serve as an adjunct to pre-operative biopsy, enhancing diagnostic confidence and potentially guiding more appropriate surgical margins, particularly in borderline or ambiguous cases, and if the model continues to improve, could potentially one day supplant the need for pre-operative biopsy completely.

Moreover, one of the key goals of integrating AI into clinical workflows is to reduce the number of steps toward diagnosis and treatment. If the accuracy improves, AutoML has the potential to decrease reliance on invasive procedures like biopsies, particularly in settings where biopsy is challenging or risky. In this context, AI could not only augment diagnostic accuracy, but also simplify and streamline clinical decision-making.

What key imaging features did the AI learn to distinguish between IP, IP-SCC, and other diagnoses?

As this study utilized the Google Vertex AutoML image classification model, which functions as a "black box" deep learning algorithm, it does not provide explicit, human-interpretable feature maps or ranked imaging features used during training. In prior studies, human beings have used factors such as bone erosion or destruction, irregular or infiltrative tumor borders, heterogeneous enhancement patterns, extension beyond the sinonasal cavity (e.g., into orbit or intracranial space), and loss of smooth remodeling or lobulated appearance typical of IP, but we do not know if these are the same patterns the platform has used, as the platform is designed to autonomously learn complex spatial and textural patterns from large datasets without requiring manual feature engineering or annotation. While this approach provides excellent performance, we acknowledge that lack of explainability is a limitation of current AutoML systems. As a future step, we aim to integrate model interpretability techniques, such as Grad-CAM or saliency mapping, to better understand the features influencing classification and to align them more directly with radiological findings.

In terms of impact, the paper is likely to influence both ENT surgeons and pathologists, particularly in how treatment decisions are made without relying on traditional pathological diagnoses. The innovative nature of the findings could also open the door for future research on AI-assisted imaging in the diagnosis of other sinonasal and skull base tumors.

Thank you for recognizing the importance of our study!

Reviewer #2 (Remarks to the Author):

The manuscript "Utilizing a Publicly Accessible Automated Machine Learning Platform to Enable Diagnostic Before Tumor Surgery" presents an innovative research proposal that aims to use a publicly available platform, Google Vertex AI, and pre-operative CT imaging of patients from nineteen 116 separate institutions with IP or IP-SCC to identify malignant transformation of a benign tumor. The authors' choice of this engaging topic and innovative research proposal is highly commendable. However, to further enhance the manuscript's potential for publication, the following suggestions should be considered:

1. The authors should review the keywords using MeSH terms. If not, please replace them as necessary.

Thank you for this suggestion. We have now revised the keywords to align with appropriate MeSH terms.

Keywords: Artificial Intelligence, Deep Learning, Machine Learning, Image Interpretation, Computer-Assisted, Inverted Papilloma, Carcinoma, Squamous Cell, Paranasal Sinus Neoplasms, Multicenter Studies as Topic, Neoplasms

2. The authors are encouraged to ensure the consulted references are correctly added in the introduction section, per the journal's guidelines.

They should also carefully explain the meaning of each abbreviation used in the manuscript.

Thank you for your important suggestion.

We have carefully reviewed the manuscript and ensured that the following abbreviations are now clearly defined upon first use:

- **CT** – Computed Tomography
- **AutoML** – Automated Machine Learning
- **DICOM** – Digital Imaging and Communications in Medicine
- **ms** – Milliseconds
- **AUC** – Area Under the Curve
- **F1 Score** – Harmonic Mean of Precision and Recall
- **TRIPOD** – Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis
- **2D** – Two-Dimensional

3. **The authors might modify lines 160 to 163 from page 6 from "Our study aims to harness AI technology via an automated machine learning (AutoML) algorithm to develop a prediction model to differentiate between IP and IP-SCC, with an eventual aspiration to raise the level of diagnosis and treatment of these tumors, and many more benign tumors with the same potential for malignant transformation, in underserved areas around the globe" to "Our study aims to harness AI technology via an automated machine learning (AutoML) algorithm to develop a prediction model to differentiate between IP and IP-SCC to increase the accurate diagnosis and treatment of these lesions."**

Thank you for this suggestion. We have modified lines 160 to 163 on page 6 as recommended.

4. The authors should list the names of the 19 institutions worldwide participating in the study in a Table or supplemental data. They should also provide the number of IRB-approved protocols in that file. In addition, the authors should incorporate lines 166 to 168 at the end of the dataset section. Then, they should add the number of patients from each academic center and establish the inclusion and exclusión criteria followed.

We have added a supplemental Table with the names of institutions involved, the IRB or otherwise internationally named institutional ethics committees protocol numbers, and the inclusion and exclusion criteria. We have left out the number of patients included at each institution, as some institutions included a small enough number of patients that this would lead to the publication of identifiable data which is against privacy laws.

5. The autos should add the algorithm and code used for the analysis in the methods section.

Thank you for your suggestion. This study utilized the Google Cloud Vertex AI AutoML image classification platform, where model training and analysis were conducted using its built-in proprietary architecture. The specific algorithms and source code are not accessible or customizable by users and therefore cannot be explicitly provided. However, we have clarified this in the Methods section, including a description of the training setup, dataset configuration, node hours, region used, and the evaluation metrics generated by the platform. We also added the Project ID (cogent-sweep-424404-f4), which contains metadata retained within the Vertex AI environment for reproducibility. Additionally, we have acknowledged this as a limitation of using commercial AutoML tools, where transparency in algorithmic decision-making and full control over model architecture are restricted.

6. The authors should set a statistical analysis section.

Thank you for your suggestion. Now, we have created a dedicated **Statistical Analysis** section in the Methods, as detailed below.

Statistical Analysis

Model performance was evaluated using metrics automatically generated by the Google Cloud Vertex AI AutoML image classification platform. These included area under the precision-recall curve (AUPRC), sensitivity, specificity, accuracy, precision (positive predictive value), negative predictive value (NPV), and the F1 score (harmonic mean of precision and recall). Confusion matrices were used to derive true positives, true negatives, false positives, and false negatives from test set predictions. No manual statistical testing was performed, as the model was evaluated entirely using the internal validation and test sets managed by the AutoML framework. Reporting follows the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) guidelines.[7]

7. The authors should organize the CT images in Figure 1 into coronal, sagittal, and axial views.

8. The authors should briefly explain Figures 2 and 3.

Thank you for this suggestion. We have revised the figure captions to provide brief but clear explanations of **Figures 2 and 3** in the manuscript.

Figure 2. (Left) **Precision–Recall Curve** showing the model's ability to balance precision and recall across all thresholds. The high values across the curve demonstrate excellent classification performance. (Right) **Precision and Recall vs. Confidence Threshold** illustrating how precision and recall change with increasing model confidence. The model achieves optimal performance at a confidence threshold around 0.5, where both precision and recall remain near peak values.

Figure 3. Confusion matrix heatmap of test classification performance for the trained model. The model correctly classified **99.7%** of inverted papilloma (IP) cases and **95.8%** of inverted papilloma with squamous cell carcinoma (IP-SCC) cases. Misclassification rates were low, with **0.3%** of IP cases incorrectly labeled as IP-SCC, and **4.2%** of IP-SCC cases misclassified as IP. These results demonstrate the model's strong diagnostic accuracy.

9. The authors are encouraged to start the discussion section with a paragraph summarizing the study's most relevant findings. This will help the authors to realize the significant impact of their work.

Thank you for this suggestion. We have added this to start the Discussion section:
In this multi-institutional study, we developed and validated an AutoML model using preoperative CT images to distinguish between IP and IP-SCC. The model demonstrated excellent performance, achieving an AUC of 99.8%, with high sensitivity (95.8%), specificity (99.7%), precision (99.2%), and an overall accuracy of 99.1%. These findings support the feasibility of using an accessible AI tool to aid in the noninvasive diagnosis of sinonasal tumors, potentially improving surgical planning and patient care, especially in settings where biopsy or advanced imaging may be limited.

10. Subsequently, the authors should compare these outcomes with similar studies worldwide to further emphasize the importance of their findings. The authors should change the syntax from lines 236 to 238 on page 9 since it is difficult to follow. In addition, the authors should go deep into how diversity strengthens their research, which seems a contradictory idea because it is set as a limitation in a posterior paragraph.

Thank you for these suggestions. We have changed the syntax as suggested to make the meaning more clear.
We also agree the diversity of our patient population is a strength of the study – we have removed it from the limitations section and discussed the need for diversity in the training and development of AI algorithms further in our discussion section, as detailed below:

There has been significant study and discussion on the need for deep and diverse data sets that draw from populations around the world, if we are to hope to develop AI algorithms that are truly representative and thus accurate for all patients.[13] It is only in recent years that researchers have discovered that information long held as true and applied across all populations in medicine only hold true for the majority population included in prior studies, for example cardiac disease symptoms differing between male and female populations.[14] If we are to hope and expect that AI will do a better job than humans in prediction, whether in radiology or other domains, we must acknowledge that such an outcome depends on scientists and doctors feeding the highest level of data possible into these algorithms, which is heavily dependent on how truly representative that data is. The international collaboration of our study is a major strength of this research, as it allows for this diversity of included data, and our algorithm is more accurate and widely useful because of it.

Reviewer #3 (Remarks to the Author):

This paper explores the use of a publicly available deep-learning platform to distinguish between benign and malignant tumors in head CT scans, specifically focusing on the classification of sinonasal inverted papilloma versus squamous cell carcinoma.

While the methodology itself is not novel, as it relies on an existing deep-learning platform, its direct application to this specific clinical problem is noteworthy. The implementation of deep learning in this context has the potential to be transformative for clinicians, offering a powerful tool for diagnostic assistance. However, the paper falls short in several key areas.

The dataset description lacks sufficient detail, making it difficult to assess the quality and diversity of the data used for model training. Additionally, the image preprocessing steps are inadequately explained, leaving uncertainty about how the data was prepared and standardized. The model training process is described in a rather superficial manner, without enough depth to evaluate its robustness or reproducibility.

Thank you for this comment. We have expanded our description of all that was done. One of the strengths and extrapolatability of this study is the lack of need to pre-process images for an AutoML in the same way one would need to for traditional deep learning algorithms, which we have described in more detail in this section, as below. Pre-processing and standardization of images requires much more time and effort, and that is exactly what the AutoML can do for us, but of course we are then limited in our understanding of what exact portions of each image it is using to predict outcome. We have added to our description of any pre-processing or labeling of the data, as detailed below:

<u>Dataset</u>

Patients with biopsy-proven diagnoses of either IP or IP-SCC were retrospectively identified from 19 academic centers, totaling 958 cases (878 IP and 80 IP-SCC). From these, 41,099 CT scan slices were extracted, encompassing axial, coronal, and sagittal planes (Figure 1). These images were labeled based on pathology results and used to train a two-dimensional (2D) image classification model using the Google Cloud Vertex AI AutoML platform (the training process simply consists of feeding this data into the LLM and telling it what the diagnosis is related to each image). The dataset included a broad range of scanner types, with slice thicknesses ranged from 0.5 mm to 1 mm, with voxel sizes of approximately 0.5–0.6 mm × 0.5–0.6 mm, and imaging protocols varied, reflecting real-world heterogeneity. No image resizing or segmentation was performed to preserve original imaging characteristics and improve generalizability across diverse scan types.

Image processing

The extracted CT scans were anonymized and stored as de-identified Digital Imaging and Communications in Medicine (DICOM) files, which were subsequently converted to JPEG format prior to model training. All images were used in their raw form, with no preprocessing steps applied for artifact removal, noise reduction, or intensity normalization. No windowing was performed; the original intensity values were preserved. Additionally, there was no manual segmentation or annotation of tumor regions — full-frame slices, including both tumor and normal anatomy, were utilized. Labels were applied at the scan (exam) level based on final resection pathology-confirmed diagnoses of IP or IP-SCC.

To simulate real-world conditions, all axial, coronal, and sagittal slices from the full sinus CT scans — spanning from the mandible to the skull base — were included, regardless of whether a tumor was visible in a specific slice. The dataset encompassed considerable heterogeneity in scanner types, voxel sizes, imaging protocols, and slice thicknesses across 19 academic institutions. Images were not resized manually; instead, Vertex AI AutoML automatically standardized image dimensions internally during training. No data augmentation techniques (e.g., rotation, flipping, or contrast adjustment) were applied. This approach preserved the real-world variability of CT imaging and allowed the model to learn under practical clinical conditions.

Model training

The model was developed using the Google Cloud Vertex AI AutoML Vision platform for image classification. JPEG-formatted CT slices were labeled based on final resection pathology-confirmed diagnoses of IP or IP-SCC. All labeled images were uploaded to the AutoML platform, which automatically performed a random split of the dataset into training (80%), validation (10%), and test (10%) subsets prior to model training (Table 1). This ensured that each image was used exclusively in one subset, preventing overlap between training, validation, and testing phases.

Model architecture selection and hyperparameter optimization were performed automatically through the platform's proprietary neural architecture search. Training was configured for a maximum of 16 node hours, with a target prediction latency of 200–300 milliseconds. Input images were used in their original resolution without resizing. Image standardization and pre-processing were managed internally by the platform, allowing the model to accommodate variability in image dimensions and voxel intensity.

The dataset was imbalanced (878 IP vs. 80 IP-SCC cases), and Vertex AI AutoML does not support manual implementation of class weighting or resampling. The model was trained on the complete dataset without manual adjustments.

Metadata, including training configuration and evaluation metrics, was retained within the Vertex AI environment (Project ID: **cogent-sweep-424404-f4**) for reproducibility.

We have also added these methodological limitations to the limitations section of the study:

This study has several limitations, including its retrospective design and potential selection bias. The use of Google Vertex AI AutoML introduces additional constraints typical of no-code platforms—limited control over model architecture, hyperparameters, and source code—as well as reduced algorithmic transparency and customization. Furthermore, manual implementation of class weighting or resampling was not supported, resulting in model training on an imbalanced dataset (878 IP vs. 80 IP-SCC cases), which may have biased predictions toward the majority class. Although AutoML may internally address class imbalance through proprietary optimization processes, these mechanisms are not user-accessible or transparently documented. Future efforts will focus on balancing the dataset, applying weighted loss functions, and restructuring data using patient-level splitting to improve generalizability and reduce bias.

In contrast, the discussion section is well-written and provides valuable insights. However, an important issue—the high false negative rate—is not adequately addressed. This is a critical concern in clinical applications, where misclassifying malignant tumors as benign could have serious consequences. A more thorough analysis of this limitation, along with potential strategies to mitigate it, would have strengthened the paper significantly.

We appreciate this reviewer's thoughtful observation. We agree that the risk of false negatives — particularly in the context of misclassifying malignant IP-SCC as benign IP — represents a critical clinical concern. We have now added a dedicated discussion of this issue to the limitations section of the manuscript, including potential strategies to mitigate it in future work.

Despite the strong performance metrics of the model, certain limitations merit discussion. Although the model achieved a high sensitivity of 95.8%, approximately 4.2% of IP-SCC cases were misclassified as benign IP, representing false negatives. This is a critical concern in clinical practice, where missing a malignant tumor could delay oncologic referral or alter surgical management. To address this, future work will explore strategies such as ensemble learning, integration of additional modalities (e.g., MRI, clinical history, genomics), and cost-sensitive training approaches that prioritize recall for malignant cases. Additionally, incorporating a mechanism into the AutoML pipeline to favor malignant classification in cases of diagnostic uncertainty may further reduce the false negative rate. This study represents a step toward addressing these challenges, with the ultimate goal of increasing the precision, safety, and clinical utility of AI-based diagnostic tools.

Moreover, while the study highlights the potential impact of deep learning in clinical decision-making, it would have been beneficial to include a discussion on interpretability and explainability. These aspects are crucial for clinician adoption, as they provide transparency into the model's decision-making process and build trust in AI-assisted diagnoses.

Thank you for your comment. We have incorporated this issue more clearly into our discussion.

An important consideration for clinical translation is model interpretability. As a "black box" deep learning system, Google Vertex AI AutoML does not provide saliency maps, feature attribution, or attention visualizations, limiting insight into the features driving predictions. Nonetheless, the model demonstrated strong performance (sensitivity 95.8%, specificity 99.7%, AUC 99.8%), suggesting it learned meaningful radiologic patterns. Likely features associated with malignancy include bone erosion, irregular or infiltrative borders, heterogeneous enhancement, and extension beyond the sinonasal cavity—patterns that may be subtle or overlooked by the human eye. While this autonomy enables robust classification, the lack of transparency may limit clinician trust. Future work will incorporate explainable AI (XAI) tools, such as Grad-CAM, to improve understanding of model outputs and better align them with clinical reasoning.

Overall, this paper presents an interesting application of deep learning in medical imaging, but it requires more rigor in dataset description, model training details, and a deeper discussion of critical results to be fully convincing. Future iterations could greatly benefit from addressing these gaps and integrating explainability methods to enhance clinical usability.

Reviewer #4 (Remarks to the Author):

The manuscript presents a multi-institutional study that leverages Google Vertex AI, a publicly accessible automated machine learning platform, to develop a model for differentiating between benign sinonasal inverted papilloma (IP) and its malignant transformation (IP-SCC) using pre-operative CT imaging. Drawing from a dataset of 958 patients (878 with IP, 80 with IP-SCC) across 19 international institutions, the authors created an automated machine learning model that achieved impressive results: 99.8% AUC, 99.2% precision, 95.8% sensitivity, 99.7% specificity, and 99.1% overall accuracy in distinguishing between benign and malignant tumors without requiring specialized AI expertise.

Some positive aspects:
-The study demonstrates exceptional clinical value by addressing a significant diagnostic challenge in tumor surgery. The ability to accurately differentiate between benign IP and malignant IP-SCC preoperatively has important implications for surgical planning and patient counseling. The model's high accuracy (99.1%) surpasses previously published rates from human experts, suggesting potential for real clinical impact in reducing reliance on invasive biopsies that may be subject to sampling error.

-The international multi-institutional nature of the study (19 centers worldwide) and large dataset (41,099 CT scan cuts from 958 patients) represent significant methodological strengths. The diversity in patient demographics, imaging equipment, and scanning parameters enhances the model's generalizability across different clinical settings. Unlike studies using standardized or homogeneous datasets, this approach better mirrors real-world conditions where image quality and acquisition parameters vary.

-The authors effectively demonstrate that non-AI specialists can leverage publicly accessible AutoML platforms to develop highly accurate diagnostic models with minimal coding requirements. This democratization of AI tool development is particularly valuable for the medical community, as it suggests that clinicians without extensive programming skills can still harness the power of AI for diagnosis, potentially accelerating clinical implementation and adoption.

Some shortcomings of the manuscript:
-The study population exhibits a significant class imbalance between benign IP (n=878) and malignant IP-SCC (n=80) cases. While the authors acknowledge this limitation, they could more thoroughly discuss how this imbalance might impact model performance metrics, particularly sensitivity for detecting the minority class. A more detailed explanation of techniques employed to mitigate this imbalance would strengthen the methodology.

Thank you for highlighting this important point. We agree that class imbalance and its potential impact on sensitivity—particularly for detecting the minority class (IP-SCC)—are critical concerns. We have now addressed this issue in two dedicated sections of the revised discussion:

Despite the strong performance metrics of the model, certain limitations merit discussion. Although the model achieved a high sensitivity of 95.8%, approximately 4.2% of IP-SCC cases were misclassified as benign IP, representing false negatives. This is a critical concern in clinical practice, where missing a malignant tumor could delay oncologic referral or alter surgical management. To address this, future work will explore strategies such as ensemble learning, integration of additional modalities (e.g., MRI, clinical history, genomics), and cost-sensitive training approaches that prioritize recall for malignant cases. Additionally, incorporating a mechanism into the AutoML pipeline to favor malignant classification in cases of diagnostic uncertainty may further reduce the false negative rate. This study represents a step toward addressing these challenges, with the ultimate goal of increasing the precision, safety, and clinical utility of AI-based diagnostic tools.

And:

This study has several limitations, including its retrospective design and potential selection bias. The use of Google Vertex AI AutoML introduces additional constraints typical of no-code platforms—limited control over model architecture, hyperparameters, and source code—as well as reduced algorithmic transparency and customization. Furthermore, manual implementation of class weighting or resampling was not supported, resulting in model training on an imbalanced dataset (878 IP vs. 80 IP-SCC cases), which may have biased predictions toward the majority class which is IP. Although AutoML may internally address class imbalance through proprietary optimization processes, these mechanisms are not user-accessible or transparently documented. Future efforts will focus on balancing the dataset, applying weighted loss functions, and restructuring data using patient-level splitting to improve generalizability and reduce bias.

-The manuscript lacks sufficient technical details about the specific algorithms employed by the Google Vertex AI platform. While the authors highlight the ease of use of AutoML, a deeper explanation of the underlying architecture, feature extraction methods, and model selection processes would enhance reproducibility and provide more transparency about the "black box" nature of the system.

Thank you for this valuable comment. We appreciate the importance of providing transparency and technical clarity, particularly when using a "black box" platform like Google Vertex AI AutoML. In our revised manuscript, we have clarified that one of the core limitations of AutoML is the lack of user access to specific algorithmic, including the model architecture, feature extraction process, and hyperparameter tuning methods. As a no-code platform, Google Vertex AI AutoML performs these tasks automatically through proprietary neural architecture search and internal optimization pipelines, which are not disclosed to users.

To enhance reproducibility and transparency, we have now included the following details in the Model Training section:

Model architecture selection and hyperparameter optimization were performed automatically through the platform's proprietary neural architecture search. Training was configured for a maximum of 16 node hours, with a target prediction latency of 200–300 milliseconds. Input images were used in their original resolution without resizing. Image standardization and pre-processing were managed internally by the platform, allowing the model to accommodate variability in image dimensions and voxel intensity.

The dataset was imbalanced (878 IP vs. 80 IP-SCC cases), and Vertex AI AutoML does not support manual implementation of class weighting or resampling. The model was trained on the complete dataset without manual adjustments.

Metadata, including training configuration and evaluation metrics, was retained within the Vertex AI environment (Project ID: **cogent-sweep-424404-f4**) for reproducibility.

While we are constrained by the platform's limited algorithmic transparency, we have emphasized this limitation in both the Methods and Discussion sections and proposed future work incorporating more interpretable and customizable AI frameworks to overcome this constraint.

-The retrospective design introduces potential selection bias. The authors should

provide more information about how patients were selected at each institution and discuss whether cases were consecutive or selected based on specific criteria, which could impact the generalizability of the results.

Thank you for pointing out this important potential bias so that we can better clarify. We have added this to the Discussion section:

As in any retrospective study, limitations regarding potential selection bias and lack of ability to control for confounders exist. However, having each institution simply include all patients with IP or IP-SCC tumors seen within the prior ten years, if all necessary imaging and data points were available, protected against selection bias as much as possible.

-The validation methodology appears to rely on internal validation (splitting the existing dataset), rather than true external validation on an entirely separate cohort. Given the multi-institutional nature of the study, the authors missed an opportunity to perform more rigorous validation by holding out entire institutions for testing rather than just randomly selected images.

Thank you for your thoughtful comment. We acknowledge that external validation using a completely independent dataset would provide a more rigorous assessment of the model's generalizability. In this study, we employed internal validation through random splitting of our multi-institutional dataset (80% training, 10% validation, 10% testing), as performed automatically by the Google Vertex AI AutoML platform. This approach ensured that each image was used exclusively in one subset and avoided data leakage between training and evaluation.

However, we agree that internal validation does not fully substitute for external validation. We have added this point to the Limitations section of the manuscript and emphasized that future work will focus on validating the model on an entirely independent dataset to better evaluate its real-world applicability across different clinical settings.

-The manuscript does not adequately address potential confounding variables that might influence model performance. For instance, tumor size, location, calcification patterns, or other radiological features could serve as spurious correlates that the model might exploit rather than true discriminative features of malignancy.

Thank you for your insightful comment. We agree that confounding variables such as tumor size, location, or calcification patterns can pose a risk in radiologic AI studies by potentially introducing spurious correlations that the model may mistakenly interpret as markers of malignancy.

However, in our study, we sought to minimize this risk through several key design choices:

All cases were labeled strictly based on final pathology (IP vs. IP-SCC), rather than on radiologic appearance, reducing the risk of circular reasoning or subjective feature bias during labeling.

We included all axial, sagittal, and coronal CT slices from full sinus scans**,** regardless of imaging characteristics. This ensured that the model was trained on a wide, heterogeneous range of anatomical and imaging features, both malignant and benign.

The model was not guided or biased by manually selected imaging features. Instead, it autonomously learned patterns from full-frame images across thousands of examples, making it less likely to rely on isolated or non-generalizable cues.

While we cannot entirely rule out the influence of spurious correlates in any deep learning model—particularly with black-box systems like AutoML—we believe that our approach, combining pathology-confirmed labels with large-scale, unsegmented, and diverse input data, significantly mitigates this concern. Future studies may further explore this by incorporating interpretable AI tools and external validation cohorts. We have included this all in our discussion.

-It is very important for the manuscript to propose that doctors with few engineering technical backgrounds use publicly available AI tools, and the related methods are also worth studying. It is suggested that the author can cite more similar papers, such as: : AI-Generated Content Enhanced Computer-Aided Diagnosis Model for Thyroid Nodules: A ChatGPT-Style Assistant, arXiv, 2024, doi.org/10.48550/arXiv.2402.02401

Thank you for this suggestion, we have added this point to our discussion and added this reference:
Finally, although using AutoML models in the clinical setting can introduce apprehension and hesitancy in physicians, it is imperative that physicians without engineering or technical coding background begin familiarizing themselves with these types of widely available tools, as they will only improve in accuracy over time, and those unfamiliar or unwilling to adapt will find themselves and their patients at a significant diagnostic disadvantage.[15]

-While the authors briefly mention affordability challenges of hosting the model on Google Vertex Cloud, they should expand on the practical implementation barriers. A more detailed cost-benefit analysis and discussion of alternative deployment strategies would enhance the translational aspect of the research.

Thank you for noting this significant challenge in translation. We definitely agree that affordability and deployment logistics are important considerations for translational applicability. We have expanded the discussion to address the practical implementation barriers, including cost-related challenges, and suggested alternative strategies to enhance the accessibility of the model in diverse clinical settings:

In addition to technical limitations, practical barriers to implementation also warrant consideration. Hosting and running models on commercial cloud-based platforms such as Google Vertex AI incurs recurring infrastructure costs, including compute resources, storage, and maintenance. These expenses pose significant challenges for widespread transmission and adoption. Moreover, reliance on proprietary infrastructure may hinder scalability and long-term sustainability.

Although our initial goal was to develop a free and globally accessible diagnostic tool, the current deployment model presents financial constraints that limit broader availability. We are actively engaging with platform representatives to explore alternative solutions, such as cost-sharing arrangements or open-access hosting options, to enhance accessibility. We may eventually need to try and replicate this model with the help of our computer science and artificial intelligence expert colleagues in academia with institutional hosting support.

Future work may include a formal cost-benefit analysis comparing cloud-based deployment with on-premises or open-source alternatives. Additionally, exploring hybrid deployment models—such as edge computing or federated learning—may offer cost-effective and scalable solutions for expanding access while maintaining robust performance.

-The paper would benefit from a more thorough comparison with existing radiological criteria used by human experts to distinguish IP from IP-SCC. Including side-by-side performance metrics of radiologists using established criteria versus the AI model on the same test set would provide stronger evidence for the superior performance claimed by the authors.

Thank you for this insightful comment. We agree that a direct comparison between the AutoML model and radiologist performance using established radiological criteria would enhance the clinical interpretability and translational impact of the study.

Although a direct comparison was not made in this study, our decision to use this platform was based on a prior head to head direct comparison of a much smaller data set, noted already in our discussion. We have provided further detail about that prior study which looked at the same data set as prior human expert analysis:

An investigation then ensued to compare the previously used traditional deep learning model with an AutoML using a much smaller and different dataset than used herein for this study (an MRI data set from only two institutions). A comparison of human expert physician (radiology and otolaryngology) assessment of that same data set, which demonstrated a sensitivity of 78%, specificity of 100%, and overall accuracy of 89% to the AutoML which, with that smaller MRI dataset demonstrated a sensitivity of 75%, specificity of 92% and overall accuracy of 84%, revealed how important "experience" is for success to both humans and AI algorithms . The human experts had the benefit of years of reading thousands of prior imaging exams and applying that knowledge to the new dataset, whereas the AutoML only had the extremely small number of images to learn from.[12]

We have also added these issues to our limitation section:

To address the limitation that this much broader and diverse dataset was not compared head-to-head with human interpretation and improve transparency, future work will incorporate explainable AI (XAI) techniques—such as Grad-CAM or saliency mapping—to identify key imaging features influencing model predictions. We also plan to conduct prospective reader studies comparing radiologist and AI performance on the same dataset to better align the model's behavior with clinical decision-making and support safer, more interpretable deployment.

Please let us know of any further concerns or questions. We appreciate your time and effort in reviewing this manuscript, and hope you find these changes have significantly improved the submission.

Stanford MEDICINE
School of Medicine

June 28, 2025

Dear Reviewers,

I am writing to submit the **second requested revision** of our manuscript entitled “Utilizing a Publicly Accessible Automated Machine Learning Platform to Enable Diagnosis before Tumor Surgery” for your consideration as an Original Article publication in *Communications Medicine*.

All requested revisions and additive information have been performed and supplied.
Please see our point by point response here:

Reviewer #1 (Remarks to the Author):

One of my initial concerns pertained to the concordance between the AI model, biopsy findings, and resection specimens. The authors responded that they did not collect data from pre-resection biopsies and instead focused solely on comparing AI predictions with the final pathology of resection specimens. However, this appears inconsistent with the methodology described in the paper, where the dataset is said to include biopsy-proven cases of IP versus IPSCC. These biopsy cases were reportedly used to train the AI model.
This raises important questions: if the biopsies did not show strong concordance with the final resection diagnoses, are they truly reliable for training an AI model? Conversely, if they were concordant, then we already have a reliable diagnostic tool in the form of biopsy pathology—prompting the question of what additional value the AI model brings.
These conflicting elements make it perplex to interpret the reported high concordance between the AI model and final pathology within this internal validation. A prospective external validation study would be crucial to substantiate these findings.
In clinical practice, discrepancies between biopsy and resection diagnoses for tumors such as IP and IPSCC often arise from sampling limitations rather than histopathological interpretation. If AI can predict the likelihood of malignancy from imaging when interpreted by a low coding experience clinicians, the discussion of the paper can also focus on its other practical application which may lie in guiding targeted biopsies—helping clinicians obtain more representative samples—rather than functioning as a standalone diagnostic tool.

Our apologies for not explaining this terminology better in our first response. Perhaps this question may be arising due to specific and variable terminology used within our particular sub-specialty with regard to biopsy. Although a “biopsy” can mean taking a sample of the tumor pre-operatively, it can also be used in the context of an “excisional biopsy”, where the sampling and complete resection of the tumor is occurring at the same time. This is what happens with benign appearing sinonasal tumors more than half the time, as they are not always accessible within the nasal cavity for biopsy, and they necessitate opening of the sinuses under general anesthesia to access. If it is not accessible for an in-office biopsy or if the patient has other reasons for why this would not be a good idea (on blood thinners, etc.), due to the need for general anesthesia and scheduled OR time, the common method would be to plan for a complete resection at that same time that the biopsy is being performed, unless a frozen section specimen sent at that time came back with a reading of malignancy. In those cases, if it is still feasible to perform a complete resection in an oncologically sound fashion, with margins, etc., then the surgery would move

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forward. If it appears that would not be possible, or that a critical structure (for example, the eye) would have to be resected along with the tumor in order to ensure a completely clear margin, the patient would be woken up for discussion about these issues, further staging would occur, and possibly the patient would be taken back to the OR for complete resection at a later date.
Due to these complexities, it is only the final pathology (of the final and complete resection specimen) that is being used as our gold standard comparison to AI predictive capability. We have used "biopsy-proven" to mean final-pathology proven, because not all enrolled patients would have had a prior-to-resection biopsy, but all have a confirmational biopsy at time of final resection. We understand this may not have been clarified well enough.
We have changed the wording to reflect that the gold standard we are comparing to is the final pathology from the complete final resection specimen. We have completely removed the word "biopsy" from the Materials and Methods section and replaced it with "pathology" or "final pathology". We hope this clarifies the issue.
Thank you for your time in reviewing this paper!

Reviewer #2 (Remarks to the Author):

I appreciate the authors' effort and kindness in following the suggested modifications to their manuscript. The authors share an engaging and innovative research topic that is highly commendable. Thus, the manuscript is optimal for publication.

Thank you for your time in reviewing this paper!

Reviewer #3 (Remarks to the Author):

The paper has improved due to the comments and the authors have made significant efforts to address all comments.

Thank you for your time in reviewing this paper!

Reviewer #4 (Remarks to the Author):

I believe the authors have provided comprehensive responses to my previous review. They've effectively addressed key concerns regarding class imbalance, technical transparency, validation methodology, and implementation barriers. The manuscript now more clearly acknowledges limitations while strengthening its clinical relevance. Their explanations about mitigating selection bias and confounding variables are particularly satisfactory. While direct comparison with radiologist performance remains a future direction, the authors have outlined clear plans to address this through explainable AI techniques and prospective studies. The revisions have significantly enhanced the manuscript's methodological clarity and translational value, making a compelling case for the utility of accessible AI tools in clinical diagnosis.

Thank you for your time in reviewing this paper!

Thank you for your consideration.
Sincerely,

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Utilizing a Publicly Accessible Automated Machine Learning Platform to Enable Diagnosis before Tumor Surgery

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Running Title: AI-Powered Differentiation of Benign and Malignant Tumors on CT

Keywords

artificial intelligence, Generative Pretrained Transformer, image classification, machine learning, deep learning, inverted papilloma, cancer, sinonasal carcinoma, malignancy, tumor, multicenter study, computed tomography

Financial disclosures

NDA: consultant/advisory board for: 3-D Matrix, Acclarent, Optinose
JAA: consultant for Medtronic and Optinose, equity and consultant GlycoMira, speaker for Glaxo Smith Kline, advisory board Sanofi
RC: consultant for Optinose, Sanofi/Regeneron, and Lyra Therapeutics
MTC: consultant for SoundHealth, Korust Co.
PGC: consultant for Medtronic, speaker for Optinose and GlaxoSmithKline, research funding from Aerin Medical
CD: consultant for Myelin Healthcare
TSE: research support from Sanofi-Aventis US LLC and Regeneron Pharmaceuticals Inc.
MF: consultant for Sanofi, GlaxoSmithKline, and AstraZeneca
MG: consultant for: Medtronic, Stryker
DAG: consultant for Pocket Naloxone
PHH: consultant for: Medtronic, Stryker; Equity in SoundHealth
JVN: consultant for: Aerin Medical, SoundHealth; Equity in SpirAir Inc
JNP: consultant/advisory board for: 3-D Matrix, Acclarent, Optinose
AJP: consultant for Medtronic, Tissium, Fusetec, speaker's bureau Sequiris, GSK, Sanofi, equity in Chitogel
JR: consultant for Storz, Spirair, Equity in SoundHealth
ZMP:consultant/advisory board for : Medtronic, Dianosic, Optinose, Mediflix, ConsumerMedical; Equity in Olfera Therapeutics, SoundHealth, Wyndly

Conflicts of interest:

None relevant

Word Count: 3269

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Abstract

In benign tumors with potential for malignant transformation, sampling error during pre-operative biopsy can significantly change patient counseling and surgical planning, such as whether critical structures like the eye or brain need to be resected along with tumor. Sinonasal inverted papilloma (IP) is the most common benign soft tissue tumor of the sinuses, yet it can undergo malignant transformation to squamous cell carcinoma (IP-SCC), for which the planned surgery could be drastically different. Artificial intelligence (AI) could potentially help with this diagnostic challenge, and allow surgeons to rely on pre-operative imaging instead of invasive biopsies and possible incomplete tumor sampling to guide their plan; however most physicians do not have training in developing these models. Here we show the utilization of a publicly available platform, Google Vertex AI, and pre-operative CT imaging of patients from nineteen separate institutions with IP or IP-SCC, to identify malignant transformation of a benign tumor with excellent accuracy. Potential over-fitting of the model was strenuously tested and disproved. An AutoML model, created from a publicly available artificial intelligence tool, by physicians with little coding background, was able to differentiate benign and malignant tumors with better accuracy than previously published rates from expert otolaryngologists and neuroradiologists.

Introduction

There are dozens of benign tumor types that occur throughout the body that have the potential for malignant transformation. Adenomas, meningiomas, lipomas, fibromas, endometriomas, chordomas, and more are examples. When these tumors are adjacent to critical structures, the ability to know whether the tumor is truly benign or malignant before surgical resection, which may need to remove critical structures in order to actually save a patient's life in the setting of cancer, is paramount. Inverted papilloma (IP) is the most common benign soft tissue tumor of the sinonasal cavity. With a 15% chance of recurrence after surgical resection and a 7-10% chance of conversion to malignancy (inverted papilloma associated squamous cell carcinoma, IP-SCC), this benign tumor has been treated with the respect and care typically afforded to cancer.[1] Early identification of malignant transformation is crucial, influencing both treatment strategy and patient counseling. However, accurate pre-operative diagnosis of IP-SCC presents a significant challenge. Conventional modalities such as in-office biopsies, computed tomography (CT) scans, and magnetic resonance imaging (MRI) yield valuable results, however challenges persist, particularly for less-experienced radiologists and surgeons, in accurately diagnosing IP-SCC through these methods.

The integration of artificial intelligence (AI)-based automated medical imaging diagnosis has revolutionized diagnostic accuracy, addressing concerns related to human error. By training diverse algorithms on annotated datasets, machine learning (ML) equips them to identify patterns and features relevant to various stages of medical conditions, thereby facilitating the automatic classification of previously unseen images.

Within all fields of medicine and surgery there is growing interest in harnessing the potential of AI to enhance diagnosis and management of different pathologies, and the question of differentiating IP from its malignant transformation is one that could carry significant clinical benefit.

Previously, AI-based diagnostic systems have demonstrated increasing accuracy in distinguishing between IP and IP-SCC when incorporating MRI with multiple demographic patient and tumor factors[2,3,4], and differentiating IP from nasal polyps using endoscopic images[5,6]. Unfortunately, many communities around the world, and even within the United States, do not have direct access to such costly diagnostic tools as MRI and endoscopy. However, most communities now have access to CT.

Our study aims to harness AI technology via an automated machine learning (AutoML) algorithm to develop a prediction model to differentiate between IP and IP-SCC, with an eventual aspiration to raise the level of diagnosis and treatment of these tumors, and many more benign tumors with this same potential for malignant transformation, in underserved areas around the globe.

Methods

The study was approved by the Institutional Review Board (IRB) of 19 institutions around the world. Due to the nature of the study as a diagnostic review, the requirement for written informed consent was waived.

Dataset

Patients with IP and IP-SCC were selected through retrospective chart reviews at 19 academic centers. Pre-operative CT scans were acquired and subsequently categorized into either IP or IP-SCC groups based on pathology reports. Representative axial, coronal, and sagittal cross-sectional images from the CT scans were extracted and labeled as 2D images. These labeled images were then utilized to train a 2D image classifier employing the Google Cloud Vertex AI platform for image classification and model training.

Image processing

The extracted CT scans were anonymized and stored as de-identified DICOM files on a secured computing platform. Images were not resized, and tumor cuts were not separated from normal cuts; instead, all slices from the whole sinus CT scans, spanning from the mandible to the brain, were included. The CT scans utilized various devices that produced images with differing thicknesses, dimensions, and voxel sizes (Figure 1). Additionally, the intensity of each voxel in the images varied, testing the AI's capability to process these variations autonomously, thereby simulating practical conditions.

Model training

A single-label dataset for image classification was established to train the model. Image analysis was conducted on the 'us-central' region of Vertex AI, a Google Cloud AutoML platform. The dataset was divided into three subsets: 80% for training, 10% for validation, and 10% for testing (Table 1). No images used in the training or validation sets were used for the testing set. Specifically, each patient exam was only used in one subset. As not every image of each exam included tumor, care was taken to ensure the three subsets had complete exams instead of random slices to ascertain diagnosis. Google-managed encryption key services were employed for security. The objectives focused on achieving higher accuracy and targeting a latency of 200ms-300ms. The training process was set for a total of sixteen node hours.

Performance metrics

The AutoML platform autonomously produces metrics to evaluate the trained model's performance. It measures the model's accuracy by computing the area under the precision-recall curve (AUPRC), reflecting its proficiency in image classification. Moreover, the platform generates confusion matrices to calculate true positives, true negatives, false positives, and false negatives, as required for deriving the performance metrics of sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV), accuracy and F1 score. Reporting follows the TRIPOD guidelines.[7]

Results

Patient Cohort

The study involved a cohort comprising 958 patients, 878 individuals with benign IP and 80 with IP-SCC. Demographic details of the patients are presented in Table 2.

In this study cohort, a comprehensive collection of 41,099 CT scan cuts was analyzed, encompassing 35,216 images representing benign IP and 5,883 images depicting IP-SCC. The trained model demonstrated strong performance, achieving an AUC of 99.8%. Precision of 99.2% was observed at a confidence threshold 0.5 (Figure 2). The model exhibited a sensitivity rate of 95.8% in correctly differentiating IP-SCC cases from IP, while specificity remained high at 99.7%. (Figure 3). Overall, the model achieved an accuracy of 99.1%, with an F1 Score of 97%, underscoring its efficacy in discerning between IP and IP-SCC. With such strong results, care was taken to double and triple check against over-fitting of the model, but the results held up to this scrutiny.

Discussion

The findings of our study align with the growing body of research that underscores the potential of AI in enhancing diagnostic accuracy of tumor diagnosis, but with the recent advent and rapid evolution of AML, this accuracy and prediction capability now far surpasses anything seen prior.[8,9,10,11]

Several key studies laid the groundwork for the current investigation. Yan et al. provided evidence for the value of human experts using MRI-based radiomics in distinguishing IP from IP-SCC, achieving a high AUC with a combined model of radiomic and morphological features.[12] However, in that study the predictive value of different parameters were able to reach the high level found in this study only when sacrificing either sensitivity or specificity, but the predictive capability could not accommodate both. Following that, Liu et al. explored the use of traditional convolutional neural networks (CNNs) to differentiate IP from IP-SCC based on MRI images. However, their sensitivity and specificity were lower than the previously reported human expert capability, and also lower than what this study achieved with only CT images.[3]

With the knowledge gained from those studies and the recognition that a larger dataset would allow for greater accuracy, our study builds on these findings by utilizing an international, multi-institutional dataset of CT images, encompassing a wide range of imaging parameters and conditions. This diverse dataset contributes to the generalizability of our AI model, which demonstrated an AUC of 99.8%, precision and recall rates of 99.2%, and an overall accuracy rate of 99.1%. These metrics surpass that of the previous studies, indicating the potential for this approach to provide superior diagnostic performance. It is important to note that while this model can distinguish IP from IP-SCC, it is not “predicting” transformation, a step further which we could hope to aspire to in the future.

One of the significant strengths of our study is its international multicenter nature, involving 19 institutions from around the world and the large number of CT images with varying dimensions, thicknesses, and voxel sizes. This variability mirrors real-world clinical conditions more closely than studies that rely on standardized or homogeneous datasets. The ability of our AI model to maintain high accuracy despite differences in image quality and parameters is particularly noteworthy. In real-life clinical settings, images are often taken using different machines and protocols, leading to variability in image characteristics. As a result, our model's high performance across such a diverse dataset suggests that it is well-suited for real-world application and could potentially reduce the need for invasive procedures like biopsies and bring high-level diagnostic accuracy to communities currently lacking in this ability.

Another strength of this study that contributes to its wide applicability is the diverse dataset provided by the multi-institutional nature of the study. This diversity enhanced the model's ability to generalize across different patient populations and imaging conditions. Also, the use of a large dataset with over 41,000 CT scan slices provides a solid foundation for training and validating the AI model, reducing the likelihood of overfitting and improving the model's reliability.

Our study highlights the transformative potential of using AutoML in developing AI models. The transition to AutoML marks a pivotal shift in methodology, as evidenced by a recent comparative analysis involving Google Vertex AI (AutoML) and the traditional All-Net neural network.[13] Using the same dataset from two institutions, the AutoML model exhibited an overall accuracy rate surpassing that of the traditional All-Net model without the need for specialized graduate-level education in artificial intelligence. The AutoML models demanded only a few lines of code, allowing us to test numerous algorithms simultaneously within a brief timeframe. This capability enabled us to swiftly pinpoint promising model algorithm classes for further development, a process that is typically time-intensive in traditional machine learning. Moreover, the user-friendly nature of AutoML makes it accessible to healthcare practitioners without extensive programming skills, paving the way for wider adoption in clinical settings.

Limitations of our study include its retrospective nature, lack of control over patient diversity, and the potential impact of class imbalance on model performance. Additionally, the general limitations associated with AutoML, such as lack of control and customization, generalization concerns, overfitting, lack of transparency, feedback loops, and liability concerns, are present. To mitigate the impact of class imbalance, future studies will explore techniques such as oversampling, undersampling, or using weighted loss functions during model training. This work involves *in silico* evaluation of our AI model; future clinical validation studies are needed to evaluate its accuracy for screening or diagnosis in the clinical setting.

An additional limitation is the cost of hosting a model on the Google Vertex Cloud platform. The initial goal of this study was to develop an open access platform that physicians from around the world would be able to access and utilize to help in diagnosis of their patients, disseminating this model for the benefit of the world. We are currently in discussion with representatives of that company to determine how to make this feasible.

Ultimately, our goal is to advance the field of medical AI for improved patient care. Overall, while AutoML holds promise in revolutionizing the diagnostic process, its limitations must be carefully considered and addressed to ensure its responsible and effective implementation in clinical practice. Future research should focus on expanding the dataset, ensuring diverse patient populations, and exploring the integration of additional data types such as genomic and proteomic information.

Conclusion

A deep AML model, created from a publicly available AI tool using pre-operative CT imaging alone, identified malignant transformation of inverted papilloma with excellent accuracy. By leveraging a large, international, multi-center dataset and embracing the inherent variability in clinical imaging, we have developed a model that is reliable, widely applicable, and highly accurate. This work paves the way for broader clinical adoption of AI-based diagnostic tools across all medical specialties, potentially transforming patient care by reducing the reliance on invasive procedures and enhancing early detection and treatment planning.

None of the authors have competing interests related to this work.

Author contributions:

FH contributed to data collection, data curation, data analysis, and drafting of the manuscript.

GL contributed to data collection, data curation, and editing and revising the manuscript.

ET, AM, AY, DK, MF, LL, SAH, NA, JAA, KA, NB, MC, RC, MTC, PGC, DYC, CRC, NC, CMC, JMD, ADS, CD, DD, SE, TSE, JBF, MG, CG, SG, JWG, DAG, RJH, AH, PHH, AMI, NDK, MAK, DKL, AL, LHL, RL, CM, CM, EDM, JVN, EPH, JNP, VCP, AJP, JR, PS, RS, MS, ES, AS, AT, JHT, SXW, SKW, BAW, PJW contributed to data collection, data curation and editing and revising the manuscript.

ZMP contributed to conceptualization of study design, data curation and editing and revising the manuscript.

All authors take responsibility for the accuracy of the data presented herein.

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Figure 1: Different CT scan cuts in Coronal, Sagittal, and Axial views, each with different voxel dimensions and slice thicknesses, demonstrating the variety of images on which the model was both trained and validated.

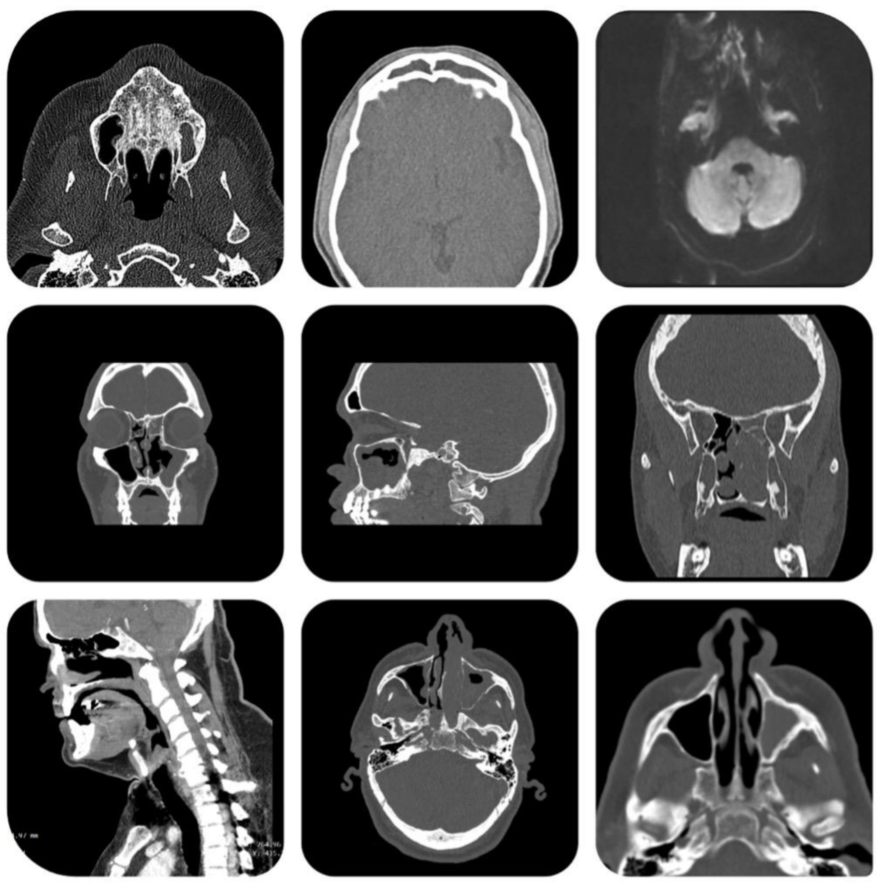

Figure 2. Precision-recall curve and Precision-recall by threshold curve for test classification performance of the trained model.

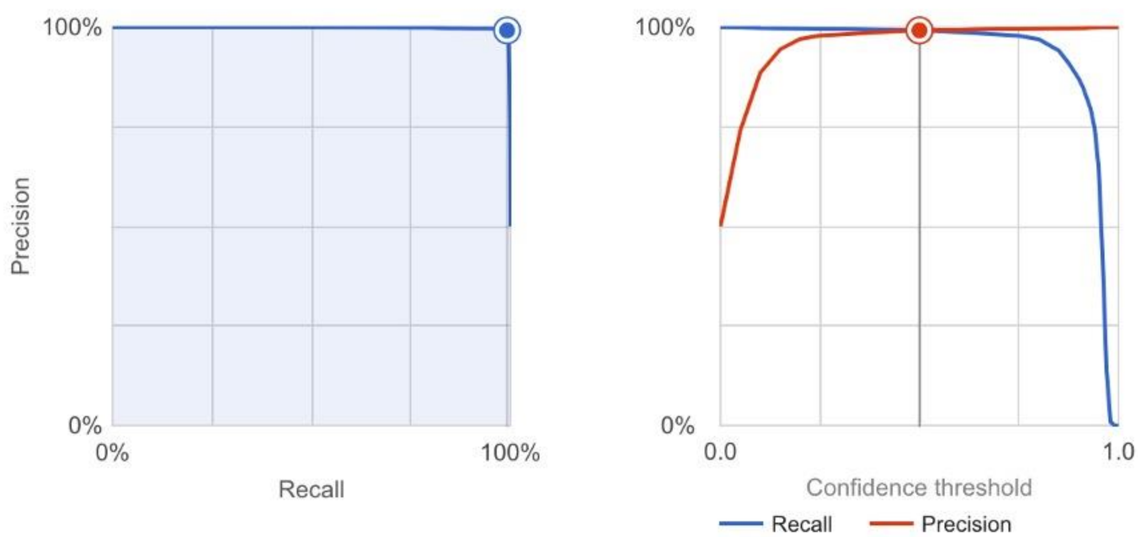

Figure 3. Confusion matrix heatmap of test classification performance for the trained model

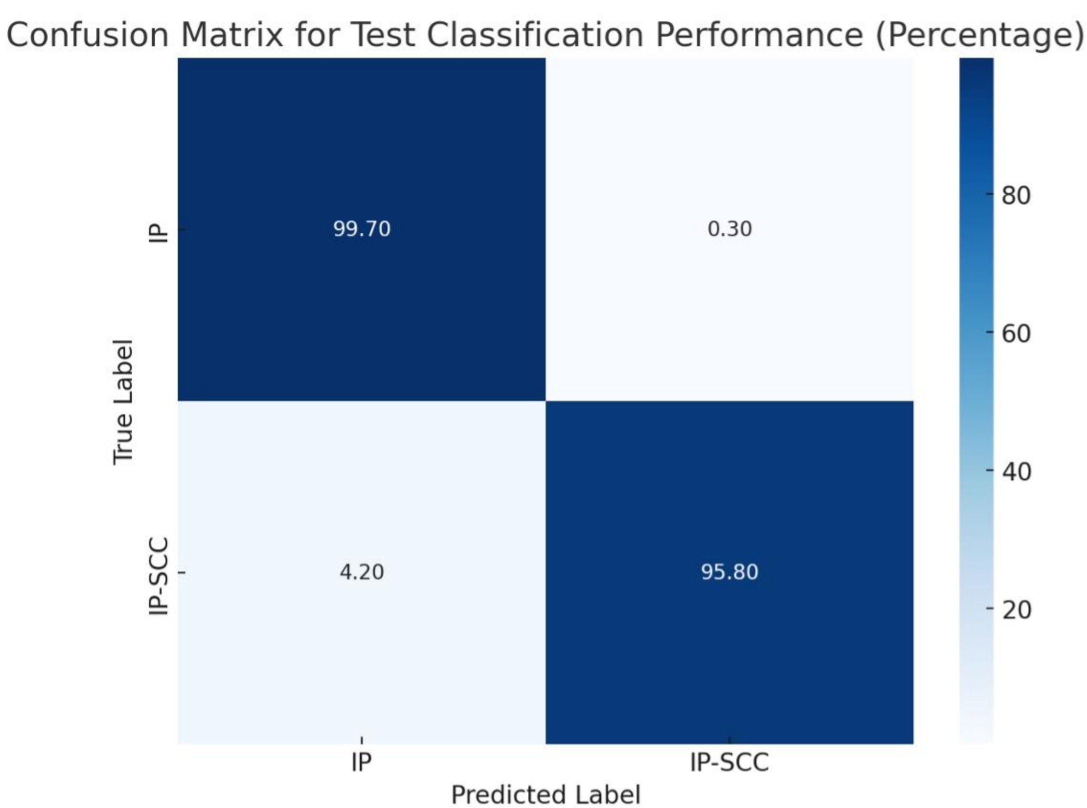

Table 1. The model split patients and CT scans into training, validation, and test groups

Labeled Dataset	Patients (n)	Total CT Scan Slices	Training set (80%)	Validation set (10%)	Testing set (10%)
IP	878	35,216	28,173	3,522	3,522
IP-SCC	80	5,883	4,706	588	589

Table 2. Patient demographic data

		IP	IP-SCC
Patients (n)		878	80
Age (mean ± SD, years)		59.26 ± 13.68	62.02 ± 12.45
Gender (n (%))	Male	596 (68)	57 (71)
	Female	282 (32)	23 (29)
Race/Ethnicity (n (%))	Asian	85 (10)	7 (9)
	White	479 (55)	25 (31)
	Hispanic/Latino	175 (20)	12 (15)
	Black	78 (9)	8 (10)
	Other	13 (1)	0 (0)
	Unknown*	48 (5)	28 (35)
Smoking History (n (%))	Yes	307 (35)	24 (30)
	No	448 (51)	29 (36)

	Unknown	123 (14)	27 (34)
Prior Sinus Surgery (n (%))	Yes	374 (43)	31 (39)
	No	504 (57)	49 (61)
History of IP recurrence (n (%))	Yes	123 (14)	17 (21)
	No	755 (86)	63 (79)

*There was missing information from some centers with regard to race/ethnicity mainly due to two factors. Many centers outside the US do not routinely record this data, and this information was also not recorded for some US patients.